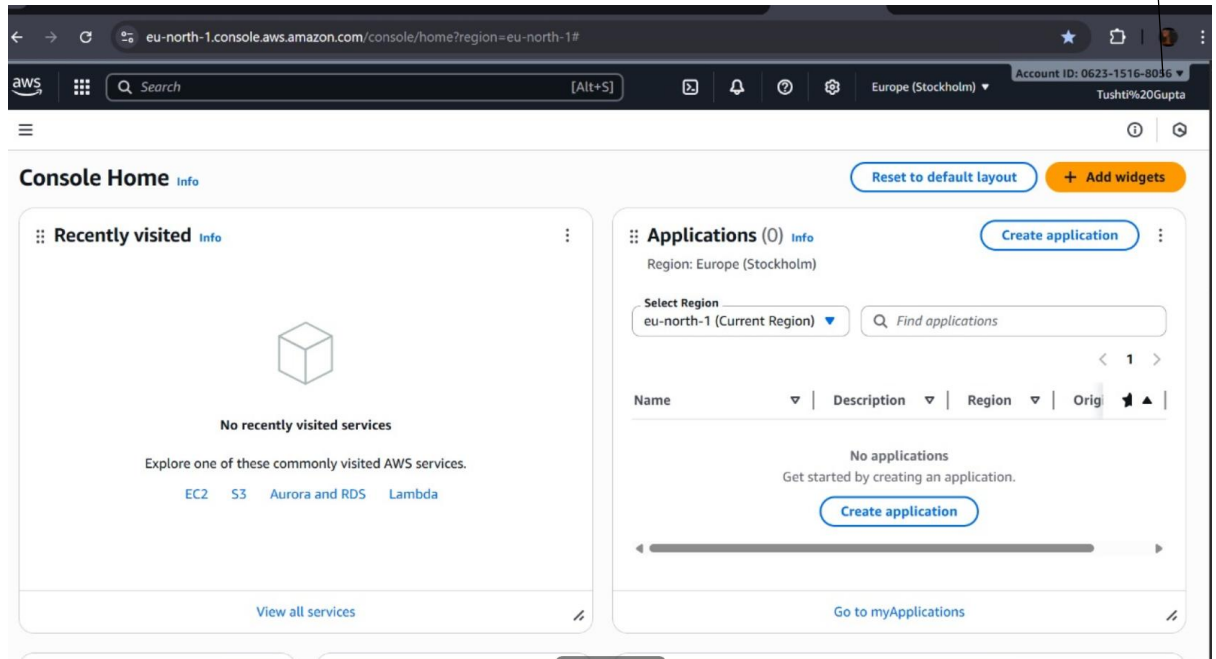


AWS Assignment-1

❖ Task 1: AWS Free Tier Account Creation

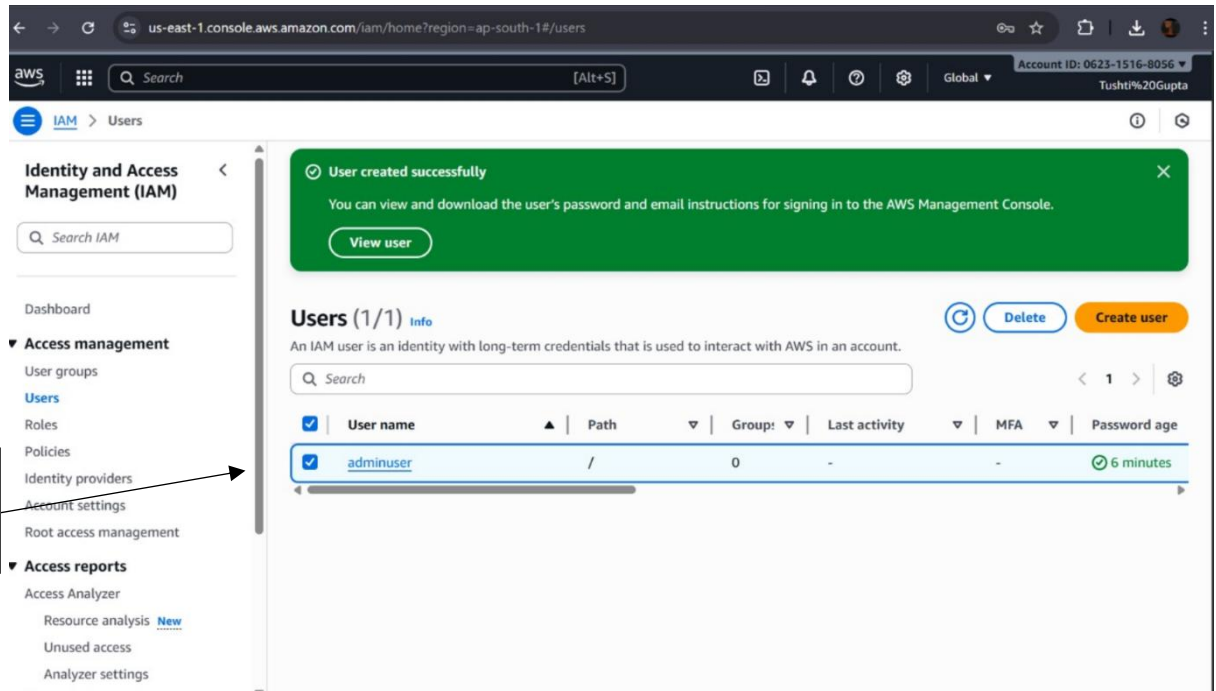
Root account



➤ Steps:

1. Go to the **AWS Management Console** → <https://aws.amazon.com/console>.
2. Click **Sign in as root user**.
3. Enter the **email address** (used when creating the account).
4. Enter the **root password**.
5. (If MFA is enabled) Enter the **MFA code**.
6. After entering the **MFA code**, root user will be created

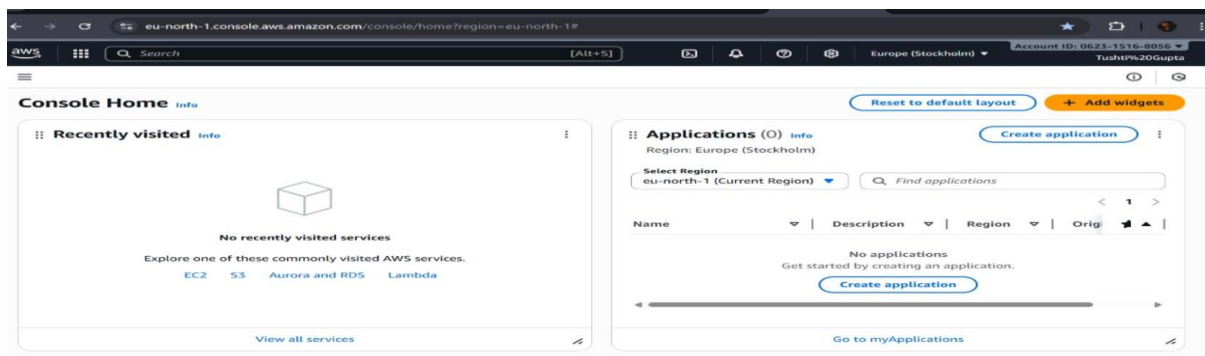
❖ Task 2: Root-Equivalent IAM User Creation



➤ Steps:

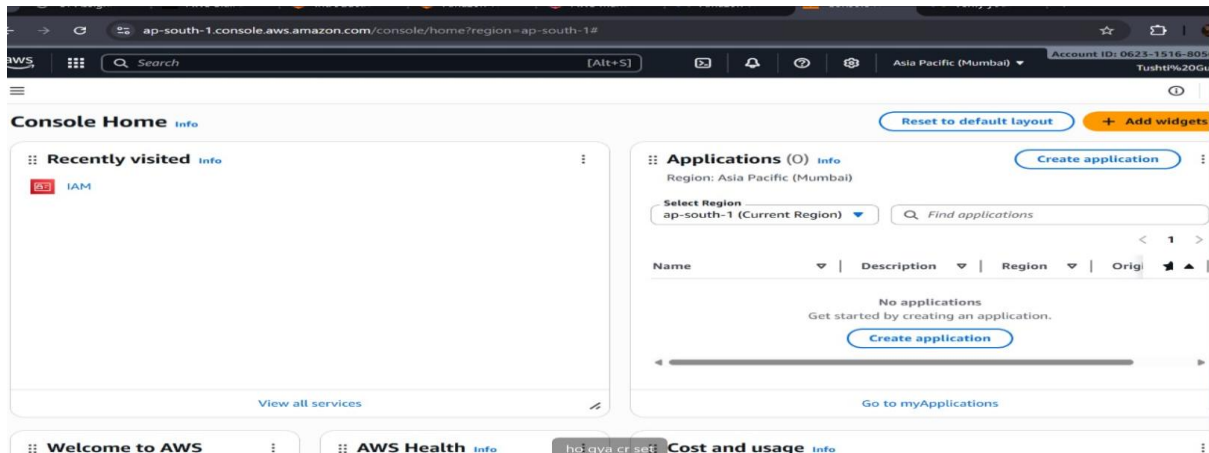
Step 1: Sign in as the Root User

- Go to the [AWS Management Console](#).
- Click “Sign in as root user”.
- Enter the **root email** and **password**.
- (If MFA is enabled, enter the MFA code.)



Step 2: Open the IAM Console

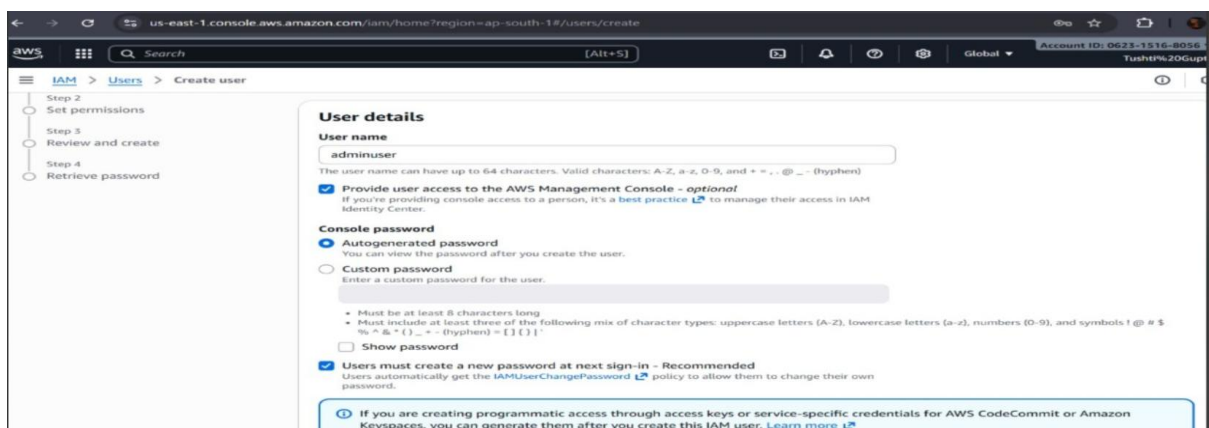
- In the AWS console search bar, type “IAM”.
- Click **IAM (Identity and Access Management)** to open it.



Step 3: Create a New IAM User

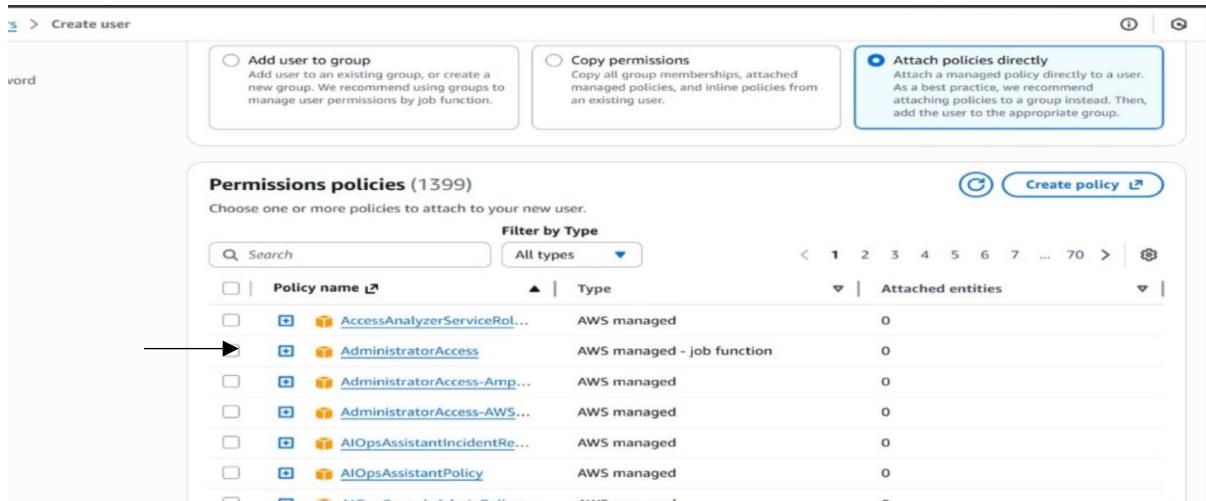
1. In the left sidebar, click **Users** → **Add users**.
2. Enter a **username** (e.g., admin-user).
3. Under **Select AWS access type**, choose:
 - ☒ **Password - AWS Management Console access** (if user needs console access)
 - ☒ **Access key - Programmatic access** (if user needs CLI or API access)

Click **Next**.



Step 4: Attach Administrator Permissions

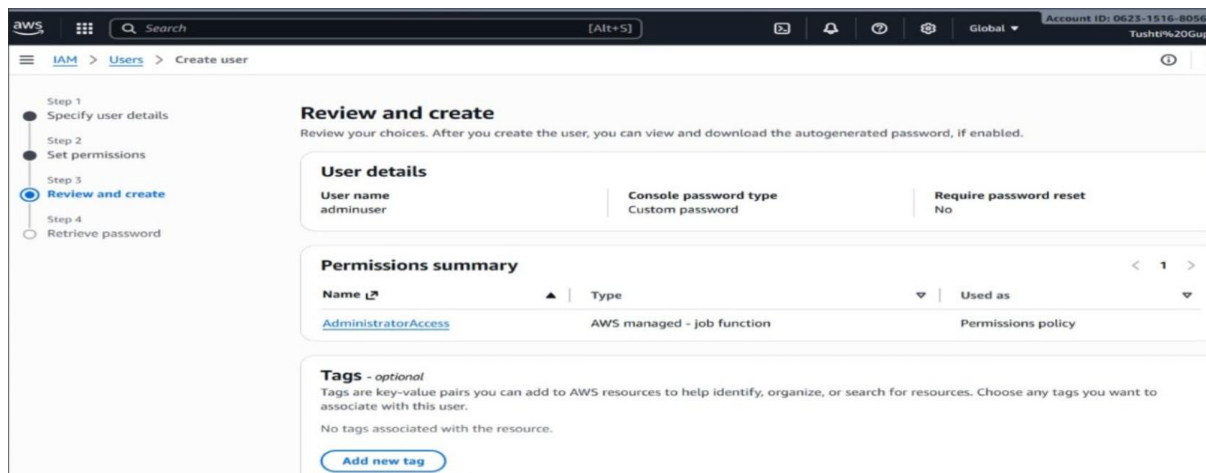
1. Choose **Attach policies directly**.
2. Search for and select **AdministratorAccess**.
 - This gives the user full permissions for all AWS services (like the root user, but still restricted by IAM).
3. Click **Next**.



Step 5: Review and Create

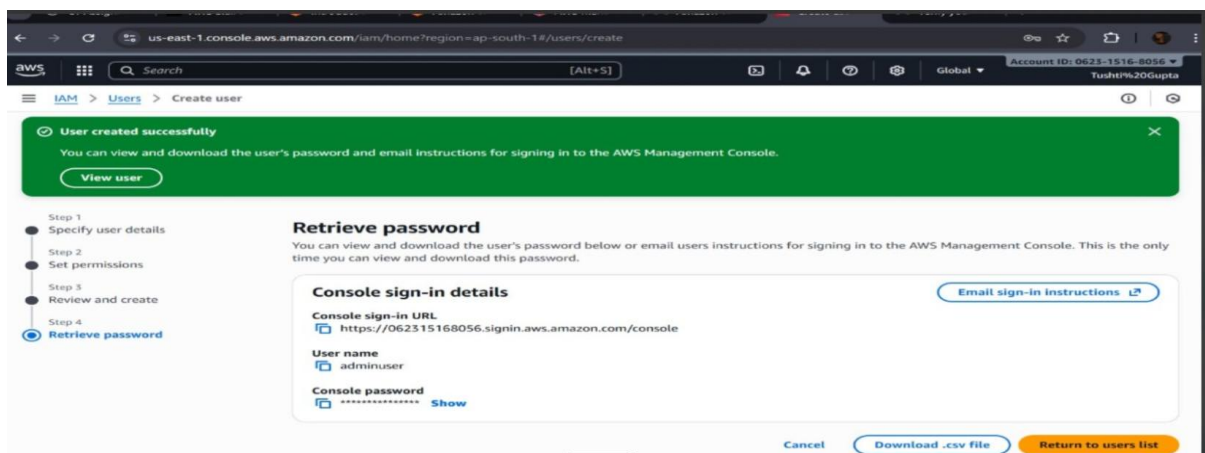
- Review all details.
- Click **Create user**.

AWS will show the **login URL**, **username**, and (if applicable) the **password** or **access keys**.

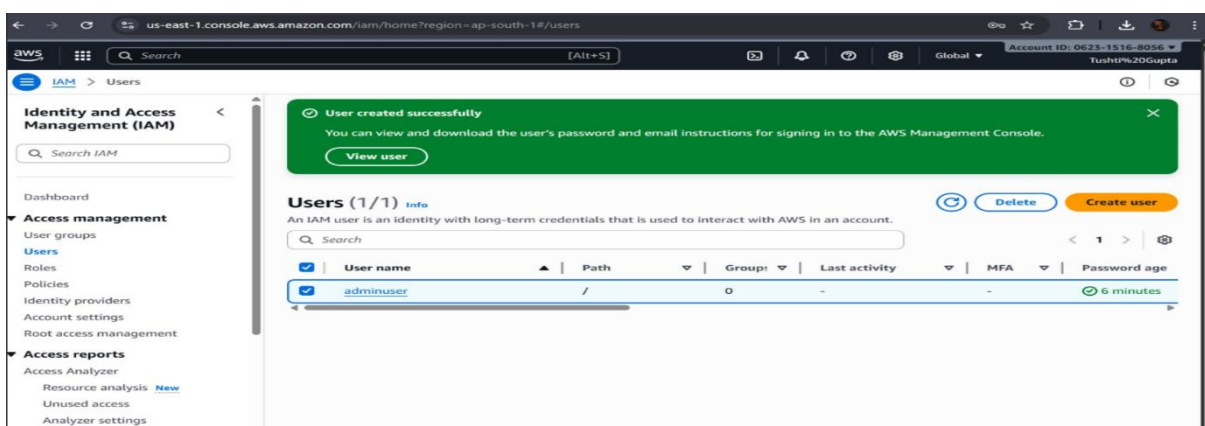


Step 6: Secure the Account

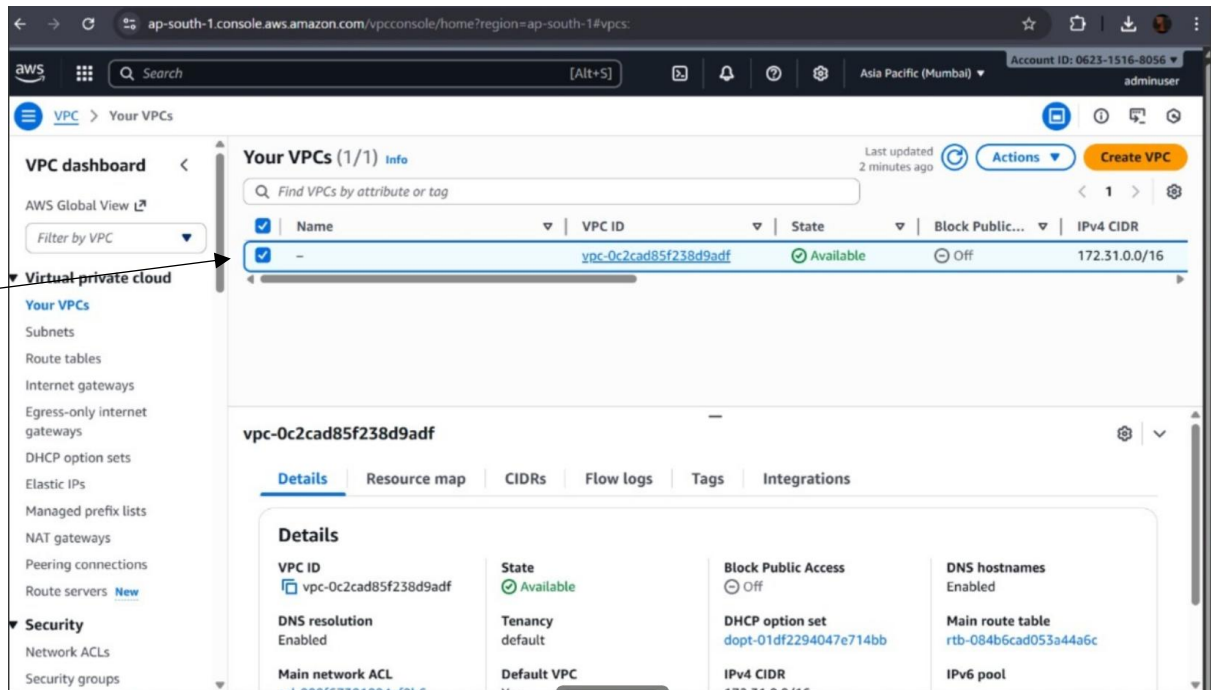
1. Sign in as the new IAM user to confirm it works.
2. Enable MFA (Multi-Factor Authentication):
 - Go to **IAM** → **Users** → [your user] → **Security credentials**.
 - Under **Multi-factor authentication (MFA)**, choose **Assign MFA device**.
 - Use an **authenticator app** (like Google Authenticator).
3. Store credentials securely (not in plaintext).



- Hence, IAM user created as adminuser



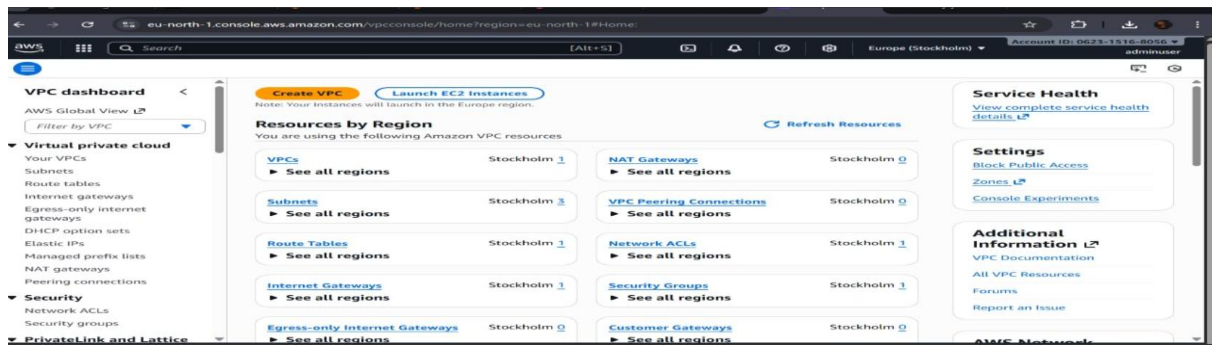
❖ Task 3: VPC Selection & Region Choice



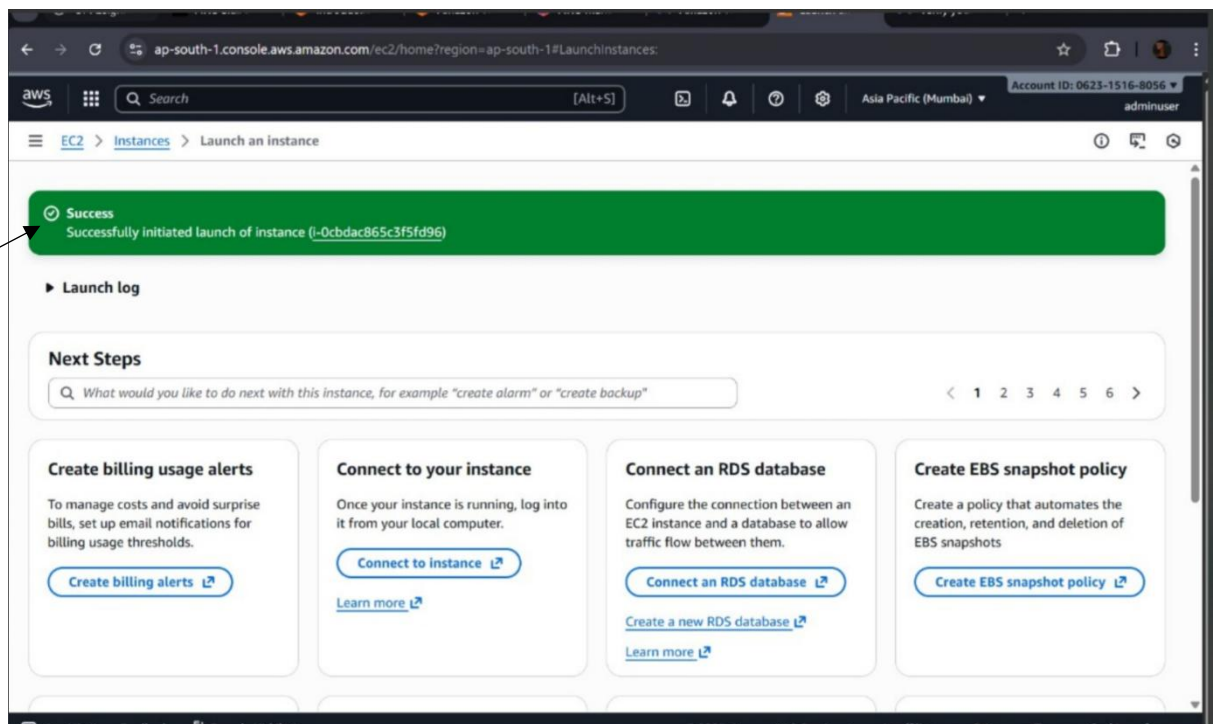
➤ Steps:

Steps to Select or Create a VPC

1. In the AWS console, go to the **VPC Dashboard** (search for “VPC”).
2. Click **Your VPCs** in the left sidebar.
3. You’ll see a **Default VPC** (it usually has “Default” in the “Name” column).
4. You can use this VPC directly for EC2, RDS, or other resources — it already has:
 - Public subnets in each AZ
 - Internet Gateway attached
 - Route tables set up



❖ Task 4: Windows Web Server EC2 Instance



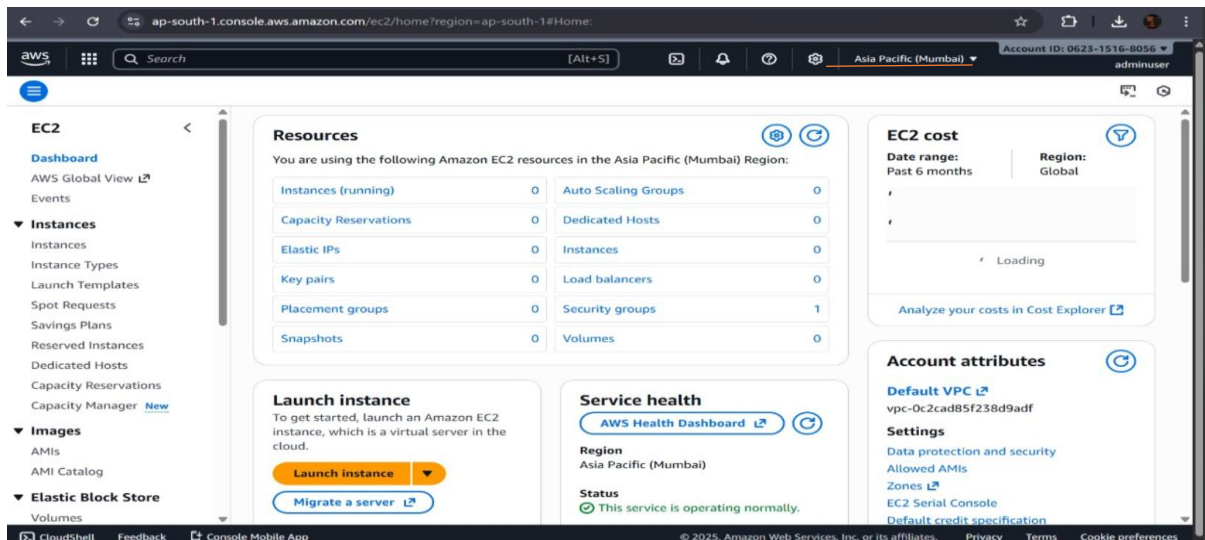
➤ Steps:

Step 1: Sign in and Select a Region

1. Go to the [AWS Management Console](#).

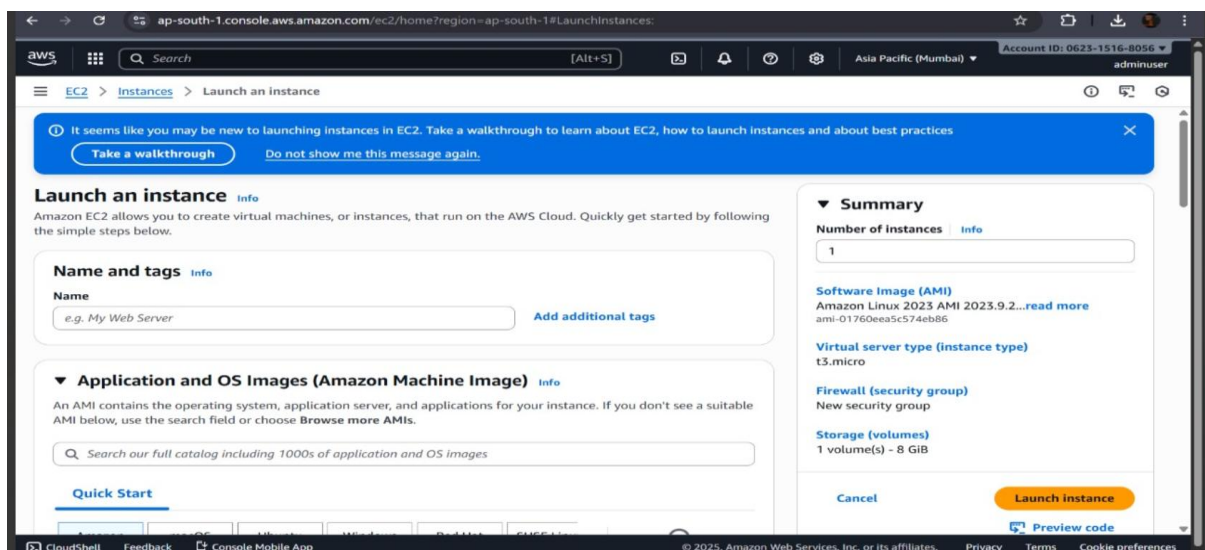
In the top-right corner, choose your **AWS Region** (e.g., Asia Pacific (Mumbai) ap-south-1).

- All resources you create will reside in this region.



Step 2: Open the EC2 Dashboard

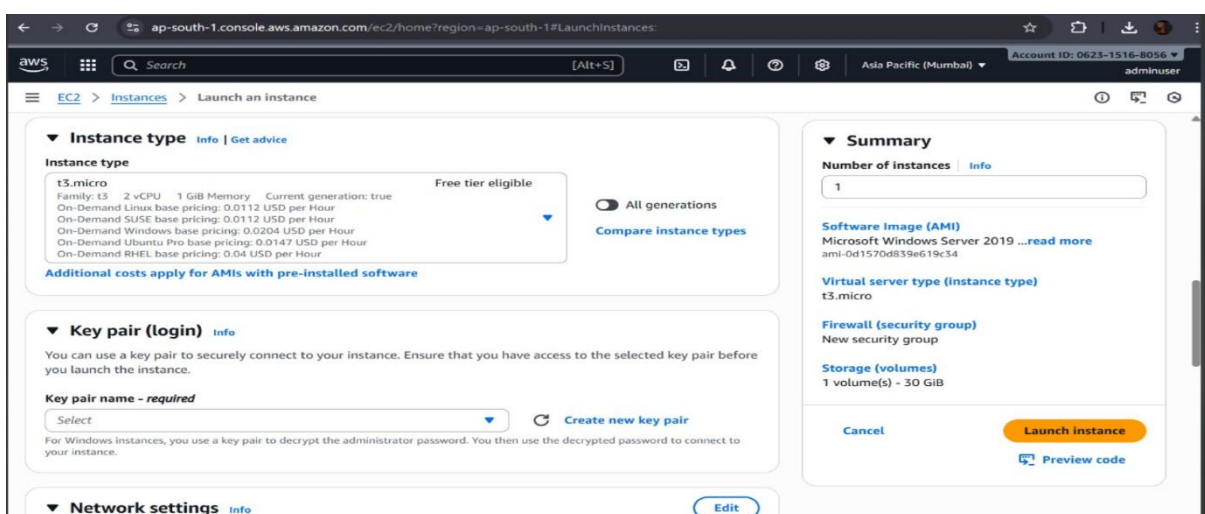
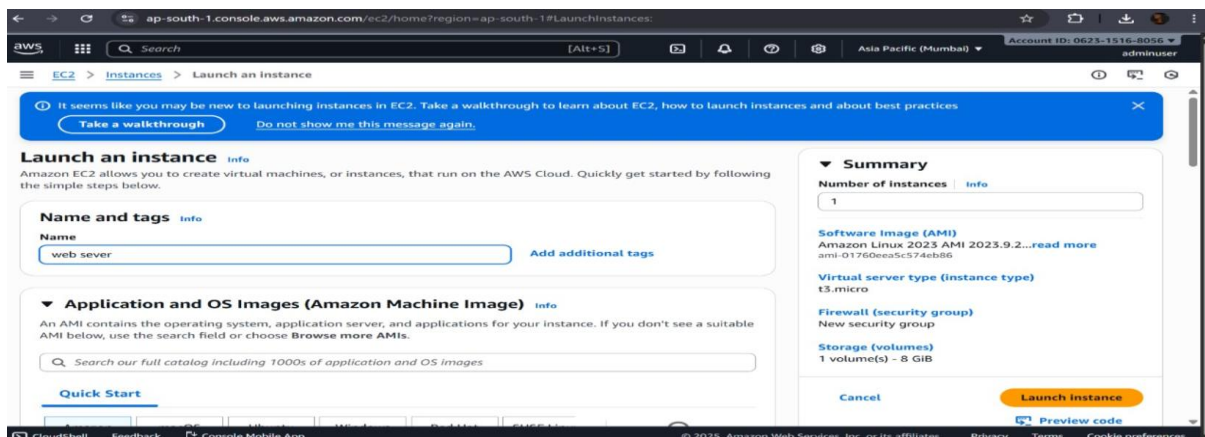
1. In the AWS Console search bar, type “EC2” and select it.
2. Click **Instances** → **Launch instances**.



Step 3: Configure Basic Details

1. **Name your instance** — e.g., Windows-WebServer.
2. **Choose an Amazon Machine Image (AMI):**
 - Scroll to find **Microsoft Windows Server 2022 Base** (or 2019/2016 if needed).
 - These AMIs come with Windows preinstalled.
3. **Choose Instance Type:**

- For basic setups, select t3.micro or t2.micro (Free Tier eligible).

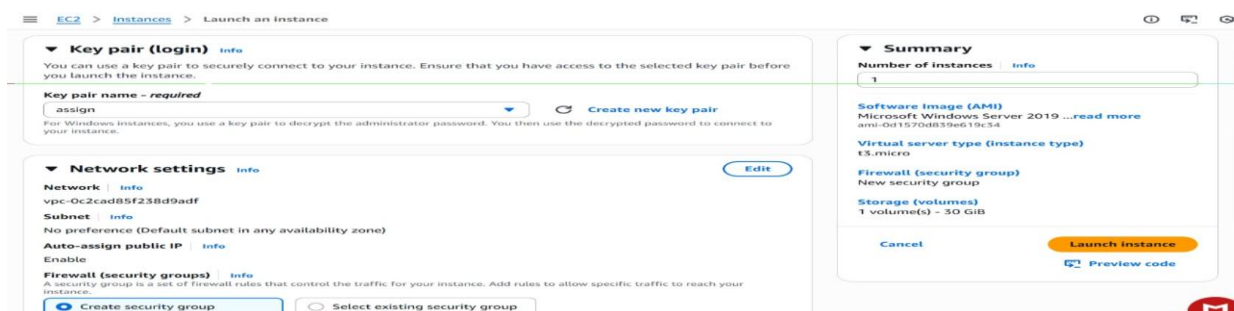
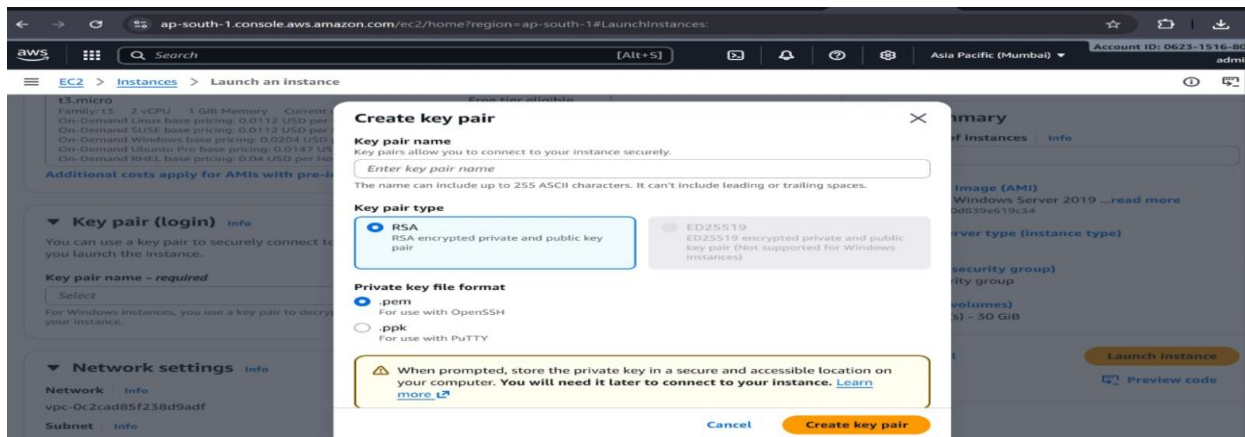


Step 4: Configure Key Pair (Login)

1. Under **Key pair (login)**, choose:

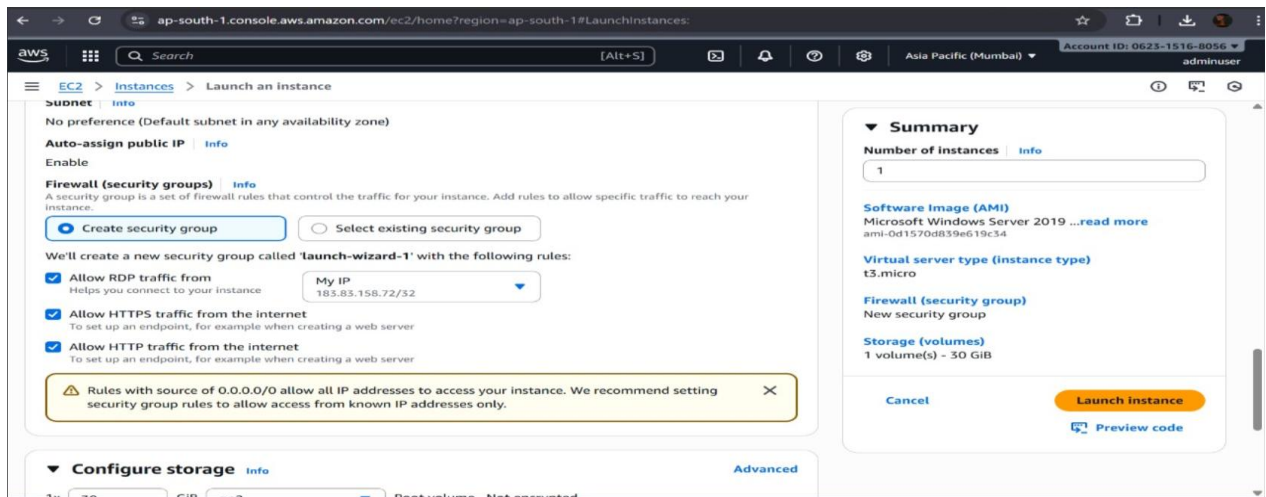
- **Create new key pair** (if you don't have one yet).
- Give it a name (e.g., WindowsKeyPair).
- Choose **Key pair type: RSA**, and **File format: .pem** (for RDP decryption).
- Click **Create key pair** — it will automatically download the .pem file to your computer.

⚠ Keep it safe — you'll need it to connect later.



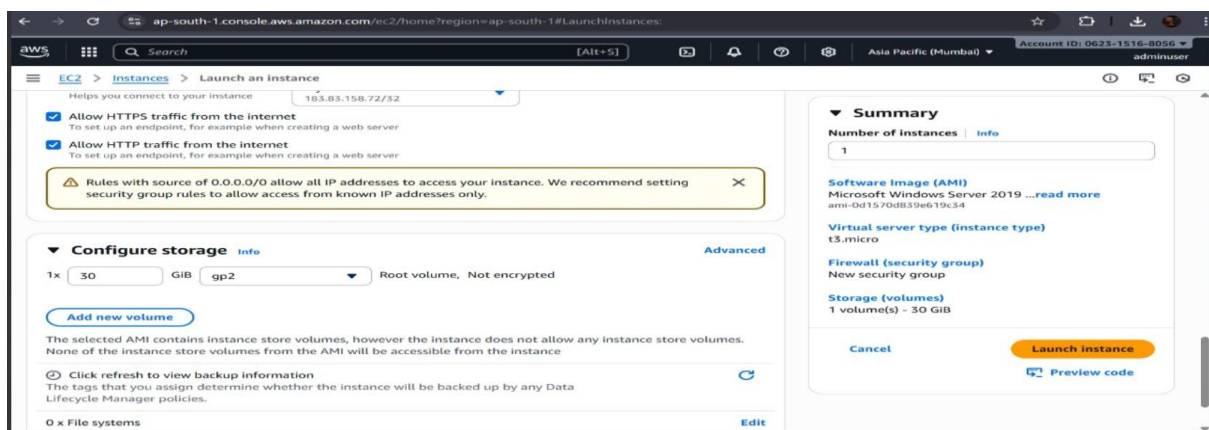
Step 5: Configure Network Settings

1. Under **Network settings**, select your **VPC** and **Subnet** (default VPC is fine).
2. Check **Auto-assign public IP** → *Enable*.
3. Under **Firewall (security group)**:
 - Choose **Create security group**.
 - Allow:
 - ☒ **RDP (Port 3389)** → Source: My IP (for remote access)
 - ☒ **HTTP (Port 80)** → Source: Anywhere (for web traffic)



Step 6: Configure Storage

- Default 30 GB is fine, or increase if needed.
- Ensure volume type is **gp3** (default).



Step 7: Launch Instance

- Review your settings and click **Launch instance**.
- Wait a few moments until the status shows **running**.

Task 5: Systems Manager Fleet Manager Setup

➤ Steps:

Step 1: Create IAM Role for Systems Manager

Fleet Manager requires the instance to have permission to communicate with SSM.

Steps:

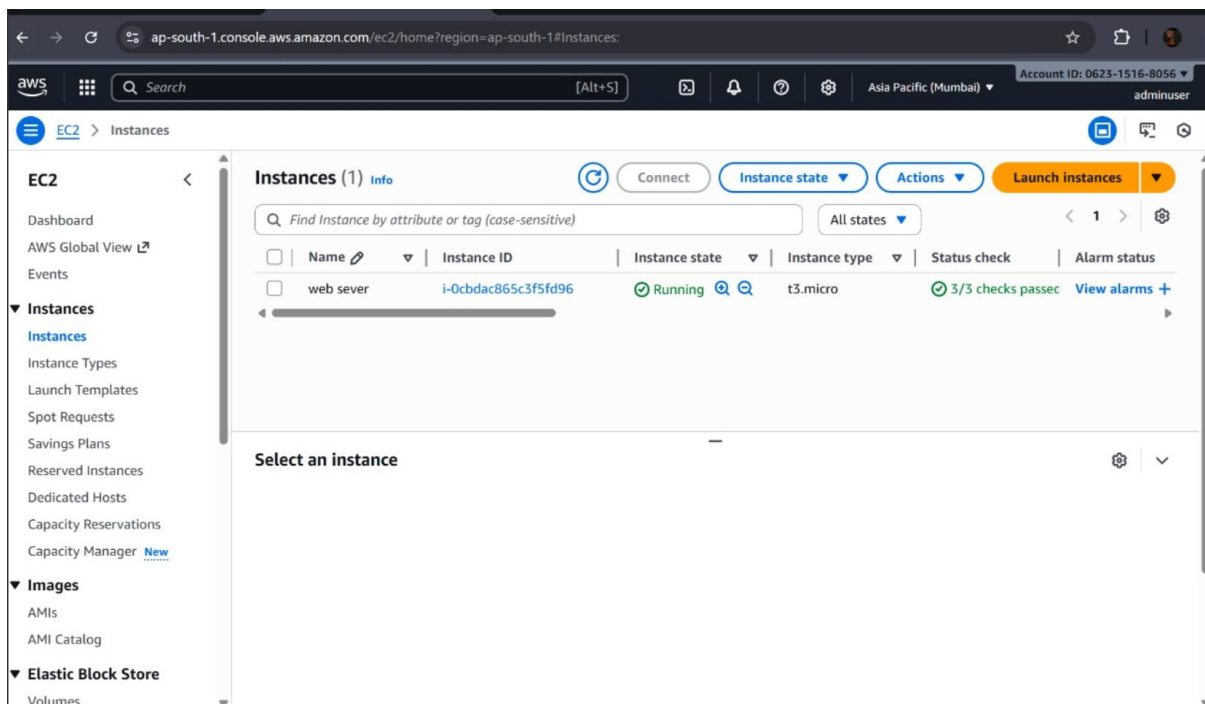
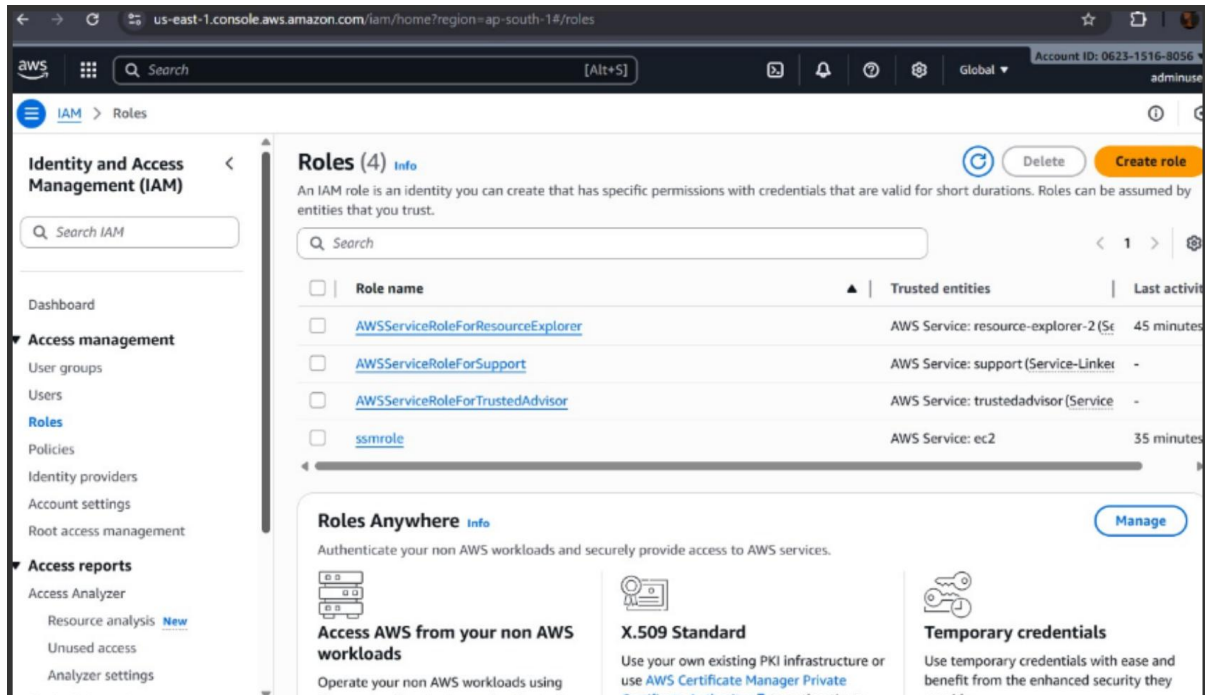
1. Go to **IAM** → **Roles** → **Create role**.
2. Under **Trusted entity type**, choose **AWS service**.
3. Choose **EC2** → Click **Next**.
4. In **Permissions policies**, search and select:
 - AmazonSSMManagedInstanceCore
5. Click **Next**, name the role:
6. SSM-Managed-Instance-Role
7. Click **Create role**.

The screenshot shows the AWS IAM console 'Create role' page, specifically Step 3: Name, review, and create. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=EC2&selectedUse...`. The page header includes the AWS logo, a search bar, and navigation icons. The left sidebar shows the navigation menu with 'IAM' > 'Roles' > 'Create role'. The main content area is titled 'Name, review, and create' and contains two sections: 'Role details' and 'Step 1: Select trusted entities'. In the 'Role details' section, the 'Role name' field is filled with 'ssmrole' and the 'Description' field contains the text 'Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.' The 'Step 1: Select trusted entities' section shows a 'Trust policy' field with a JSON snippet:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
```

Step 2: Attach IAM Role to EC2 Instance

1. Go to **EC2** → **Instances**.
2. Select your instance → **Actions** → **Security** → **Modify IAM role**.
3. Select SSM-Managed-Instance-Role → click **Update IAM role**.



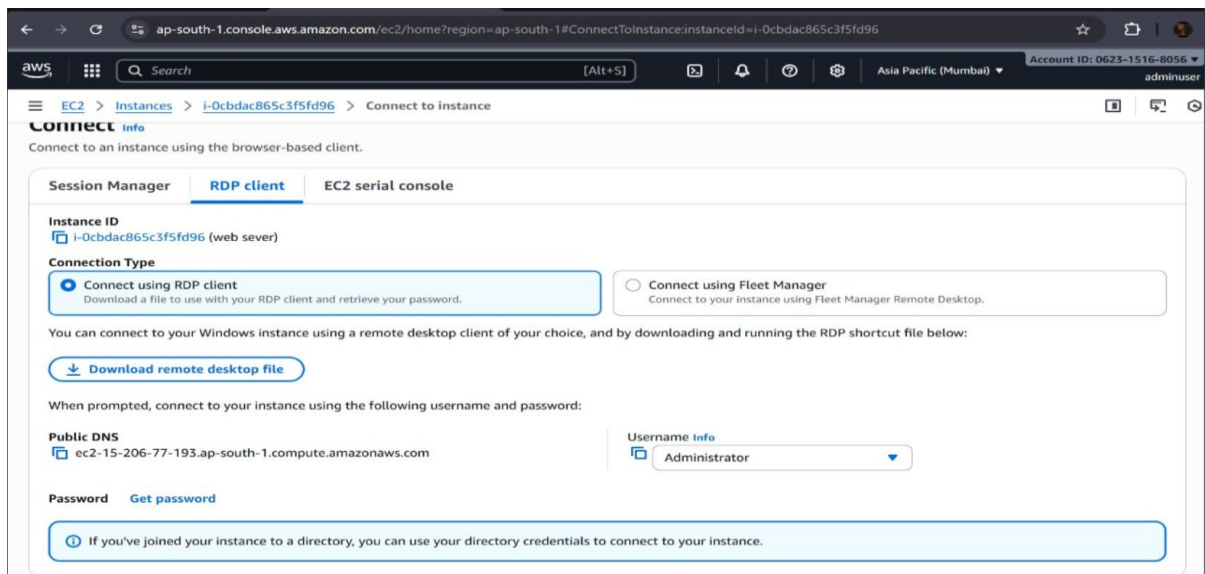
Step 3 Verify SSM Agent is Installed

For Amazon Linux 2 or Windows:

- The SSM Agent is preinstalled.

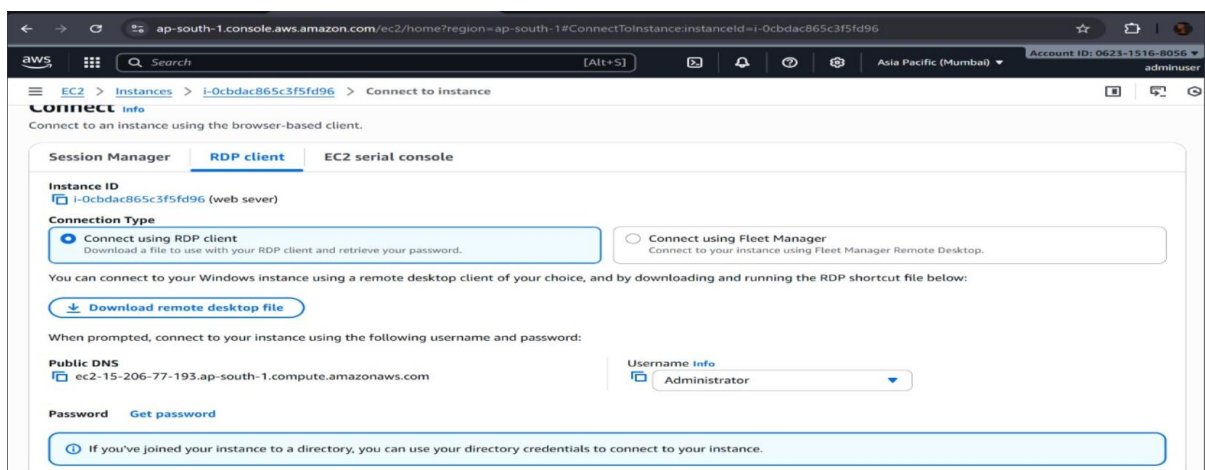
For Other OS (e.g., Ubuntu, CentOS):

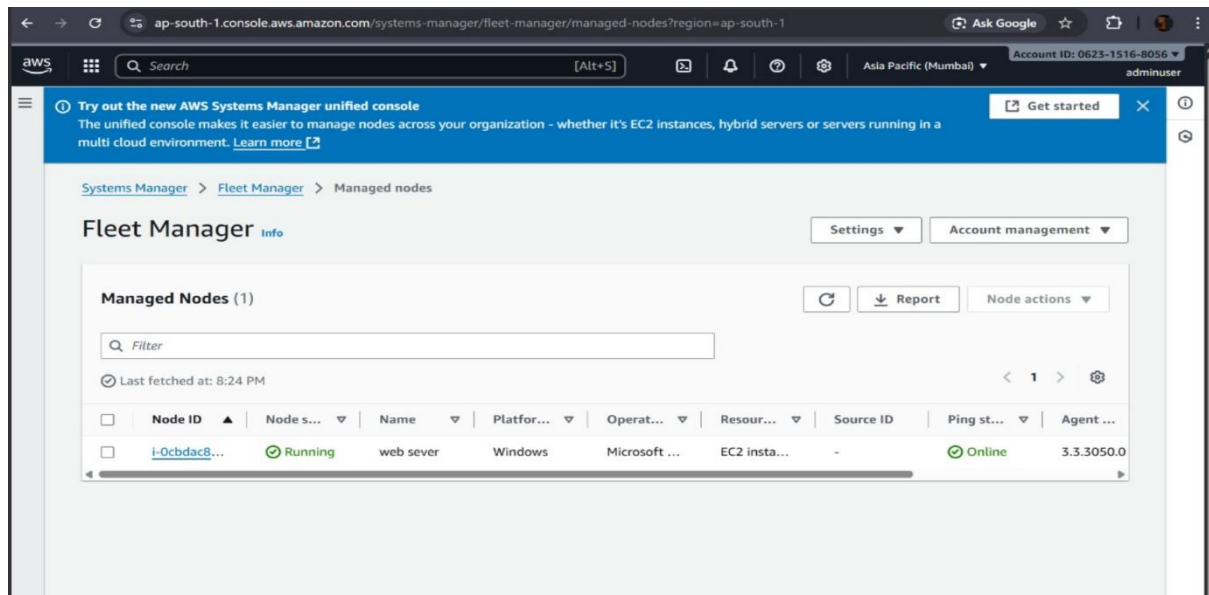
1. Connect via SSH or RDP.
2. Run these commands:



Step 4: Open Fleet Manager

1. In the AWS Console, go to **Systems Manager**.
2. In the left menu, under **Node Management**, click **Fleet Manager**.
3. You should now see your EC2 instance **listed**.
 - If it doesn't appear yet, wait a few minutes for SSM to register it.



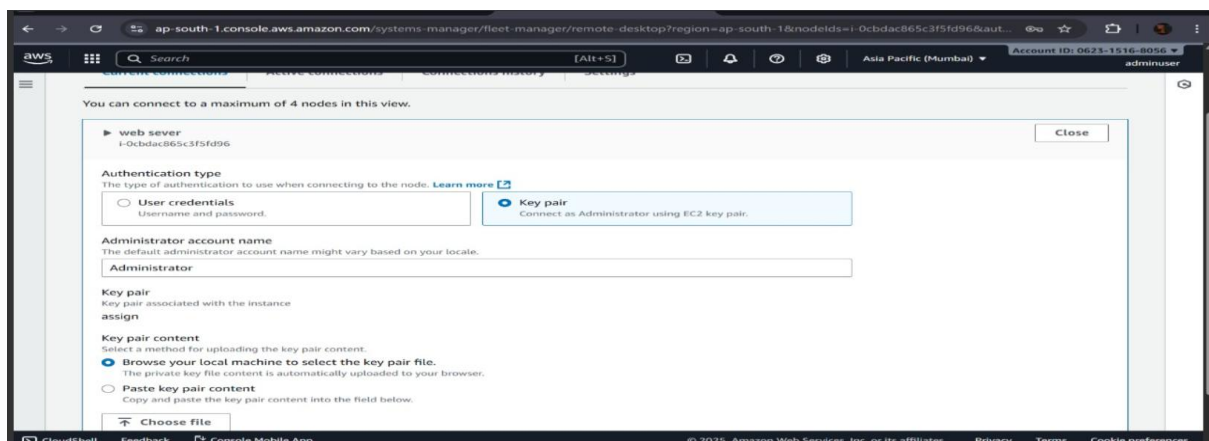


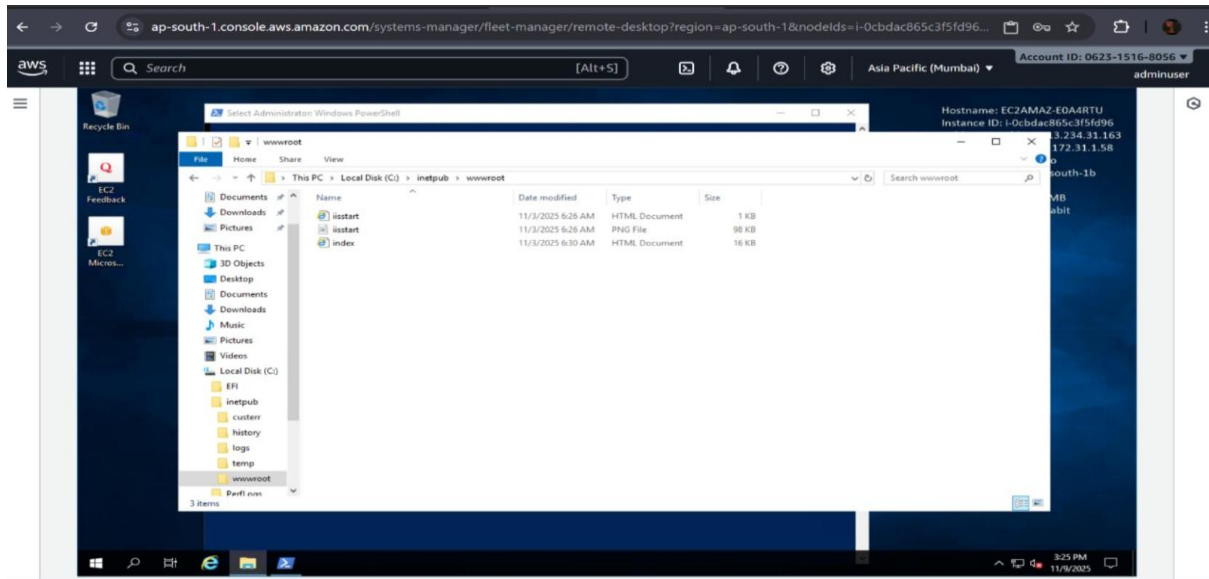
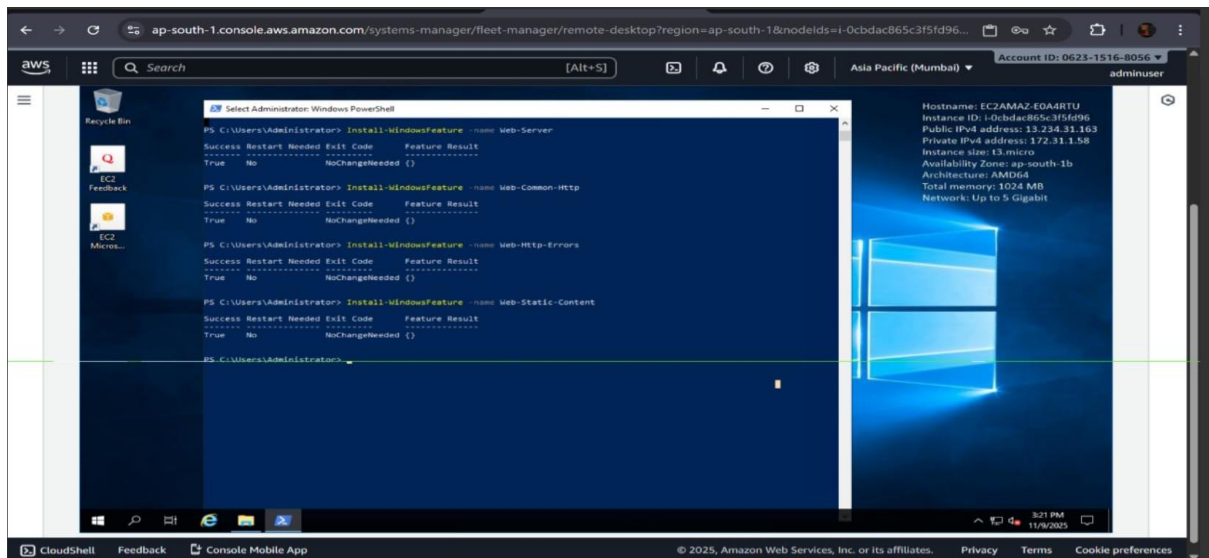
Step 5: Manage Instance via Fleet Manager

Now you can manage your EC2 instance **without RDP or SSH**:

From Fleet Manager Dashboard:

- Select your instance → click **Node actions** or instance name.
You can now:
- View system info (CPU, memory, network, OS details)
- Browse and edit files
- View event logs
- Manage users
- Run PowerShell/Command Line directly in console
- Access the Windows registry (for Windows)
- Reboot or stop instance remotely





Final Output:

